HARD GEOMETRY AND TRIGONOMETRY

Right triangles and trigonometry

03

10 questions · ~15 min

SAT QUESTION BANK 2026-05-29

Q1 GRID-IN GEOMETRY AND TRIGONOMETRY

In triangle XYZ, angle Y is a right angle, the measure of angle Z is $33°$, and the length of $\overline{YZ}$ is 26 units. If the area, in square units, of triangle XYZ can be represented by the expression $k\tan 33°$, where k is a constant, what is the value of k?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

GEOMETRY AND TRIGONOMETRY

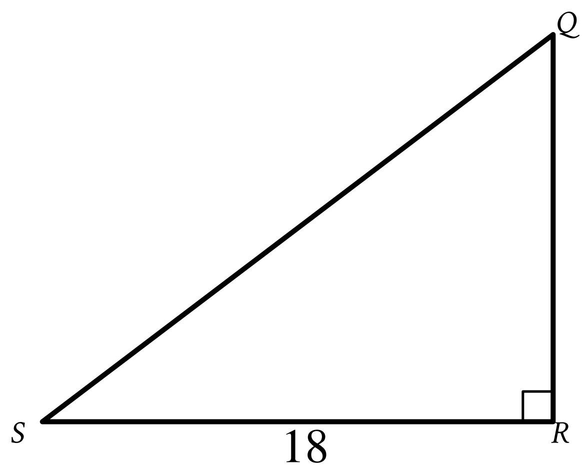
In a right triangle, the tangent of one of the two acute angles is $\frac{\sqrt{3}}{3}$. What is the tangent of the other acute angle?

A. $-\frac{\sqrt{3}}{3}$

B. $-\frac{3}{\sqrt{3}}$

C. $\frac{\sqrt{3}}{3}$

D. $\frac{3}{\sqrt{3}}$



Note: Figure not drawn to scale.

- Angle upper R is a right angle.
- The length of side upper R upper S is 18.
- A note indicates the figure is not drawn to scale.

In triangle QRS shown, $QR < RS$. Which expression represents the length of $\bar{QS}$?

A. $18\cos Q$

B. $18\sin Q$

C. $\frac{18}{\cos Q}$

D. $\frac{18}{\sin Q}$

Q4

GRID-IN GEOMETRY AND TRIGONOMETRY

The perimeter of an equilateral triangle is 624 centimeters. The height of this triangle is $k\sqrt{3}$ centimeters, where k is a constant. What is the value of k?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

In triangle XYZ, angle Z is a right angle and the length of $\bar{YZ}$ is 24 units. If $\tan X = \frac{12}{35}$, what is the perimeter, in units, of triangle XYZ?

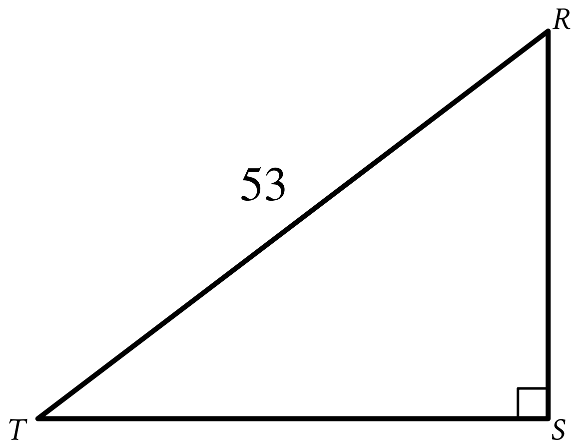
A. 188

B. 168

C. 84

D. 71

Q6 GRID-IN GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- Angle upper S is a right angle.
- The length of side upper R upper T is 53.
- A note indicates the figure is not drawn to scale.

In the triangle shown, $RS = \sqrt{105}$. What is the value of $\sin R$?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7

GEOMETRY AND TRIGONOMETRY

A square is inscribed in a circle. The radius of the circle is $\frac{20\sqrt{2}}{2}$ inches. What is the side length, in inches, of the square?

A. 20

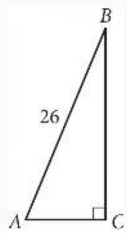
B. $\frac{20\sqrt{2}}{2}$

C. $20\sqrt{2}$

D. 40

Q8

GRID-IN GEOMETRY AND TRIGONOMETRY



Triangle ABC above is a right triangle, and $\sin(B) = \frac{5}{13}$. What is the length of side $\overline{BC}$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GRID-IN GEOMETRY AND TRIGONOMETRY

Triangle ABC is similar to triangle DEF, where angle A corresponds to angle D and angles C and F are right angles. The length of $\overline{AB}$ is 2.9 times the length of $\overline{DE}$. If $\tan A = \frac{21}{20}$, what is the value of $\sin D$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10

GEOMETRY AND TRIGONOMETRY

$$RS = 20$$

$$ST = 48$$

$$TR = 52$$

The side lengths of right triangle RST are given. Triangle RST is similar to triangle UVW, where S corresponds to V and T corresponds to W. What is the value of $\tan W$?

A. $\frac{5}{13}$

B. $\frac{5}{12}$

C. $\frac{12}{13}$

D. $\frac{12}{5}$